Modifying LeanCoP Options On the Fly

(aka "the PAAR paper")

Introduction and Research

LeanCoP inspired a large family of provers which, with a small amount of modifications, could improve the performance of the connection prover in a large set of problems [6]. The ones that inspired this work change one property of the original LeanCoP at the initial time.

LeanCoP 2.0 [5] added several options, most notably `cut` and `scut` to restrict backtracking in connection calculus and *iterative deepening* with `comp(J)`, which iteratively limits the height 1 at a time of the tree when the prover is saturated[1] until it reaches maximum height `J`. Afterwards it restarts iterative deepening from 1, but without any cut active.

This work also gathers ideas from other implementations of other provers from various different domains.

Letz and Stenz (2001) generally shows that reductions should be preferred over extensions in connections provers because they close the current subgoal immediately [3]. Generally shorter clauses tend to produce better solutions.

Zombori et al. (2021) introduces an entropy measurement to show how "confident" each policy should be: higher entropy suggests that the next step should be exploration, while lower entropy suggests that it should be exploitation [8]. Adding entropy regularisation improves training performance, and can make simpler models have better results than more complex ones.

Li et al. (2024) frames the reset choice in SAT solvers as an online policy problem [4]. It shows that recent search productivity should influence reset behaviour. Similar policies related to restarts are also presented in chapter 2 of Holden (2021).

Novel Idea

The three atoms in LeanCoP that can change the completeness of the current tree, `cut`, `scut`, and `comp(I)`, can be triggered dynamically using heuristics from the current tree rather than being chosen from the start.

It is possible to do a solution using reinforcement learning rather than heuristics. This would produce a connection prover that solves the same or fewer trees than existing provers such as rlCop [2] or learnCoP[citation soon!], thus making it redundant compared to the existing provers.

Ideally, an implementation that changes these atoms on the fly should be faster than both LeanCoP and RLCoP in a subset of TPTP problems.

Implementation

I use the existing connection prover implementation as a base for this code[7]. This allows me to use the existing code for hard things like Prolog parsing, and also to use PyCoP as a control group.

My code implemented 6 different strategies.

FirstAction *(control)* run the first legal action (PyCoP default).

LeanCoPCut *(control)* always run `cut`, `scut`, and `comp(7)`, like LeanCoP.

HeuristicSelection Choose reductions if possible, otherwise choose the extension with the shortest clause.

EntropyCut Run the first legal action, but add a cut to an extension if the Shannon entropy of the amount of atoms of this is high compared to the rest, and the prover is certain that this is the solution.

AdaptiveReset Resets the tree if the prover hits 5 dead-ends in a row.

SmartCut[2] Combines the last three strategies.

[2] Better name TBD.

Preliminary Results

I tried this on

Future Work

References

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